ic-link
by imec

WHITE PAPER

Application-specific chips (ASIC)

The benefits and how to seize them

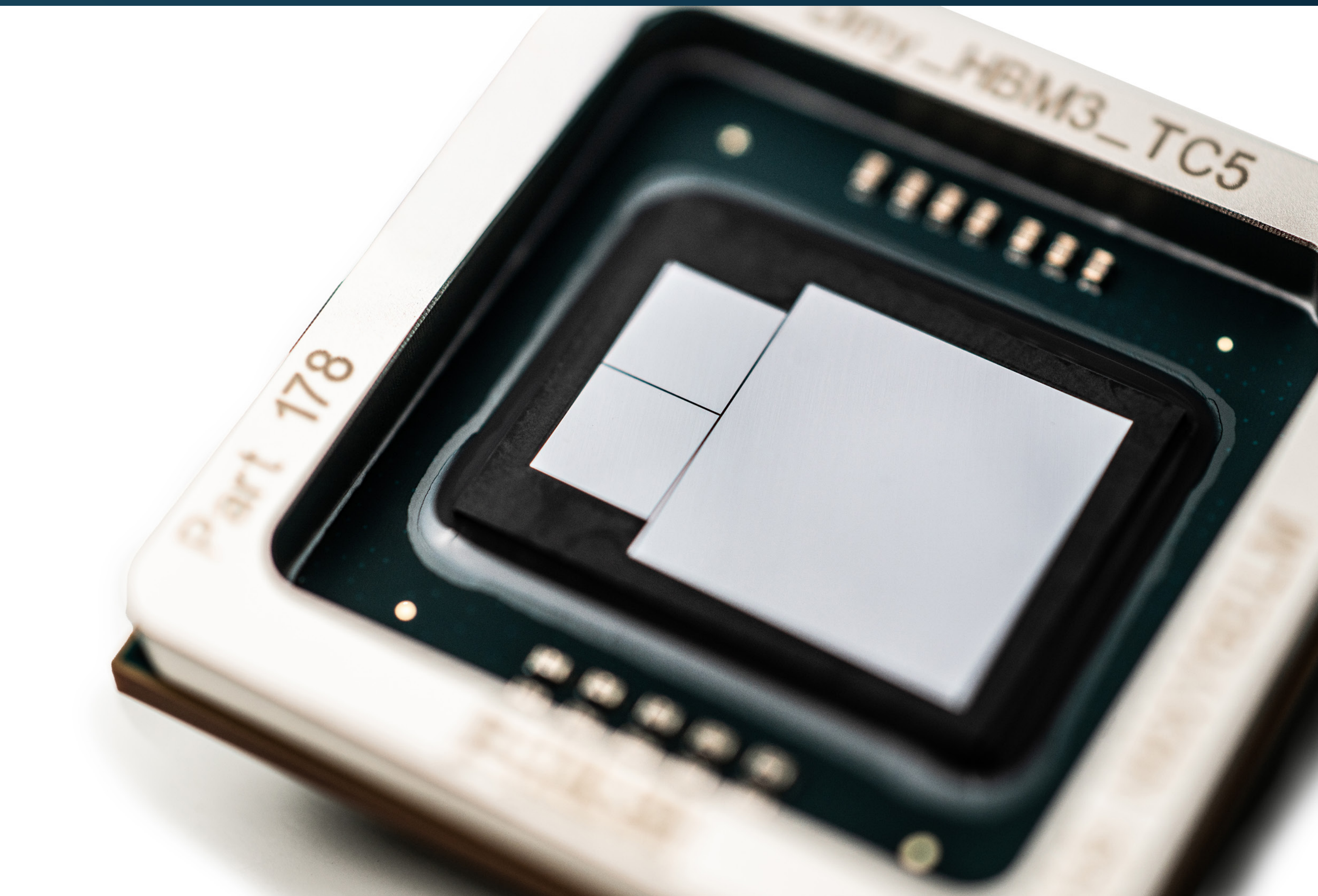
Contents

What to expect from this white paper?

This white paper is a **practical guide** for innovators looking to enhance their products with customized circuits, otherwise known as ASICs (application-specific integrated circuits). Whether you're designing medical devices, industrial systems, automotive applications, high-performance computing, data-/telecom or space solutions, an ASIC can **make your product more powerful, cost-effective**, **compact**, and **secure**, while consuming **less energy**.

Advances in technology have made ASICs more accessible and affordable than ever. This white paper provides an overview of the benefits of using an ASIC, walks you through the basics of the design and manufacturing process, and **introduces IC-Link's comprehensive services for ASICs**, from concept idea to final delivery of packaged and tested chips.

If you're a product manager, engineer, or CTO, looking to take the next step in product innovation, this guide is for you.



Introduction

From standard to full-custom chips, a choice depending on the targeted time-to-market, performance, and cost

Today's chip technology allows to integrate optical, mechanical, and electrical functionalities into products. And while determining which functionalities to incorporate is often seen as the primary focus, an equally important consideration is **the implementation approach**. Should these functionalities be realized using standard chips or fully custom chips? Read: should I use a printed circuit board (PCB) and build the system – much like assembling Lego pieces – with off-the-shelf and readily available processors, imagers, wireless transceivers, or should I design an application-specific chip, fully tailored to my application. And what about the intermediate approaches like application-specific standard products (ASSP) and field-programmable gate arrays (FPGAs)?

Standard chips offer the advantage of a low development cost, high flexibility and fast time-to-market but are not ideal in terms of performance, size and power consumption. Also in this class – and performing a bit better – are **ASSPs** which are developed for a specific application area, but not for a specific implementation. Think, for example, of MPEG decoders, PC and mobile phone chips. Another standard option is **FPGAs** which have the advantage of being programmable which allows for product updates.

ASICs are at the other end of the spectrum – the customized option, **tailored for a specific task or application**. These chips are engineered to deliver high performance with power efficiency, and enable a compact form factor. There are **full-custom ASICs**, designed from the bottom up, at transistor level, and requiring real craftmanship. And there are **semi-custom ASICs** which use predesigned building blocks and only need the higher circuit levels to be custom designed. The downside of an ASIC is that it requires upfront investment in development time (>18 months based on the complexity) and non-recurring engineering costs. But this pays off in the long run when producing significant quantities of units. Also, ASIC technology has matured significantly over the years, making it more affordable and accessible – even for companies without deep expertise in chip design or semiconductor manufacturing.

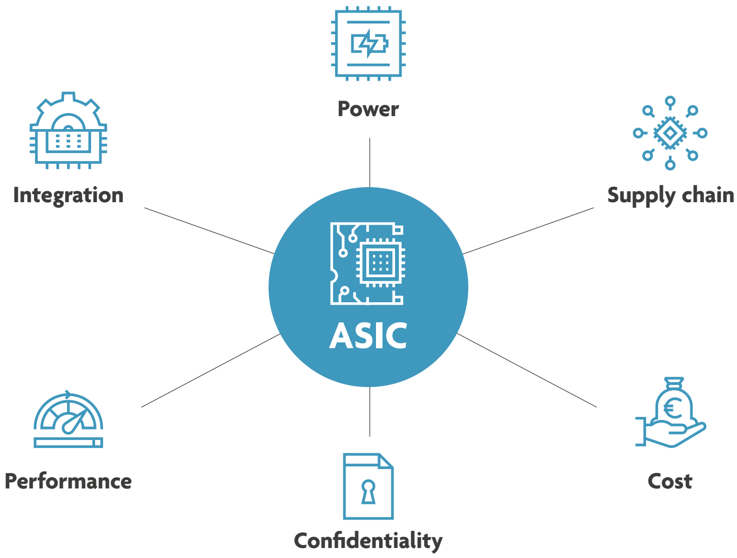
Comparison	Std IC	ASIC	Comment
Time-to-market	+++	++	Standard components are relatively ideal for prototyping and small series.
Non-Recurring expenses	+	++	Introduction cost relatively low for standard component applications.
Unit cost	+++	++++	With the optimal set-up for a given volume the unit cost for ASIC is very low especially due to no redundant functions and overhead. Large volume and small form factor are required. Large volume and smallest form factor equal lowest price for ASIC.
Scalability	++	++++	With an ASIC it is easy to increase the volume without extra effort.
Power consumption	++	++++	The design for an optimal power consumption is the real strength of ASIC technology
Form factor	+++	++++	ASIC is ideal for portable and low power applications due to its small form factor and power consumption.
Control of supply chain	++	+++	A reduced Bill of Material (BoM) will prevent many challenges of redesign as an ASIC is an integration of many components into one single chip.

This white paper will focus on the added value of an ASIC, illustrated by different projects undertaken by the IC-Link team for companies from different sectors. Additionally, it explores the common concerns and questions that arise throughout the journey from design to volume manufacturing.

IC-Link is part of the global R&D hub imec, and provides custom solutions for innovative chip manufacturing. Its offering includes end-to-end ASIC and photonic-integrated circuits, along with speciality technologies services and sensors. More information on page 13.

The key assets of an ASIC

Let's dive deeper into the main benefits of an ASIC and illustrate them with real-world products. Choosing an ASIC means opting for **a full package of advantages that are closely intertwined** and even provide benefits in areas you might not immediately expect. It's like looking at your system in a completely new way.



Cost efficiency at scale

ASICs involve substantial initial costs – the **non-recurring engineering (NRE) expenses** which cover tasks like circuit design, photomask production, package and production test development along with product qualification. However, as **production volumes** increase, the cost per unit drops significantly. And, because an ASIC enables a more compact, lighter, less complex product, this approach also saves costs in terms of inventory management, product maintenance, cooling, shipping etc.

Compactness

One of the primary advantages of ASICs lie in their ability to **combine multiple functions into a single chip and to add extra features at no extra manufacturing cost**. Through monolithic integration[1] or system-in-package approaches, the physical size of the chip and product can be reduced, and the overall circuit board design is simplified. This is especially interesting for mobile and wearable applications.

Low power consumption

By integrating multiple functions into a single chip, you minimize the energy lost in interconnections and reduce the overall power required to operate the device. This is particularly crucial for **mobile and wearable applications**, where **battery life** is a key consideration for users, as well as datacom and telecom applications where multiple units are used in one system.

High performance

As the ASIC is specifically designed for a given application, it will execute this application with the highest performance and at the lowest energy level (e.g. due to **shorter signal paths** and **optimized power management**). This is an important asset for applications where the market is very much focused on every improvement in power consumption or performance.

[1] Imec white paper "Monolithic microsystems", https://www.imec-int.com/en/what-we-offer/design-and-foundry-services/white-paper-monolithic-microsystems

IP protection

The use of an ASIC for your product protects it from being copied, serving as a **powerful obstacle to competitor entry**.

Supply chain control

With ASICs, you have a better control over your supply chain, as compared to discrete chips from multiple suppliers. This is especially important in case of component shortages or when parts become obsolete, leading to high costs because of redesigns – notably evident during the COVID-19 pandemic. In this way, the use of ASICs ensures characteristics **consistency in your product portfolio**.

ASIC projects to illustrate the added value of ASICs

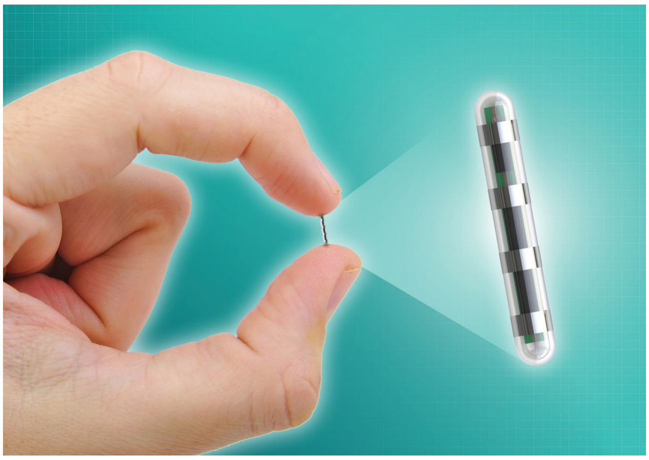
To illustrate the above-mentioned benefits of an ASIC solution, we describe here some IC-Link projects with companies from different sectors, each having their unique reason for adopting this approach.

A migraine-curing implant

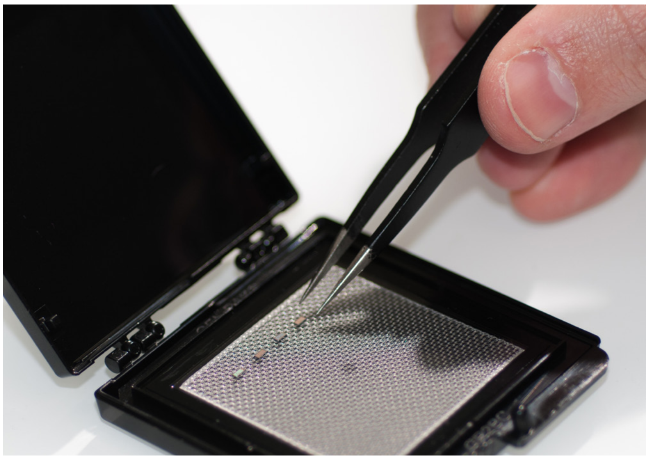
Capri-Medical is an Irish start-up that develops a miniature bioelectronics device that is implanted in the back of the neck, to stimulate the peripheral nervous system and cure chronic migraine. Current stimulation systems are implanted during a 3-hour surgical procedure. Capri-Medical wants their device to be '**injectable**', with the procedure only taking max. 20 minutes, under local anaesthesia. They chose an ASIC approach because they needed **an ultra-miniature device, that performs better than existing systems and integrates multiple functionalities, at a reasonable cost**.[2]

Smart tags for automatic identification

Wiyo is a Spanish start-up that produces smart tags, to track or interact with all kinds of objects, e.g. in hospitals, stores, industrial environments etc. The tags have no battery and no dedicated readers. They harvest power from standard 2.4 GHz Wi-Fi waves. Just a few microwatts are enough for generating power to run the smart tags. The fact that Wiyo wanted their product to work **without batteries** implied that it needed to be **extremely power efficient**. That's why they chose an ASIC approach.[3]



Capri-Medical's implantable bioelectronics device.



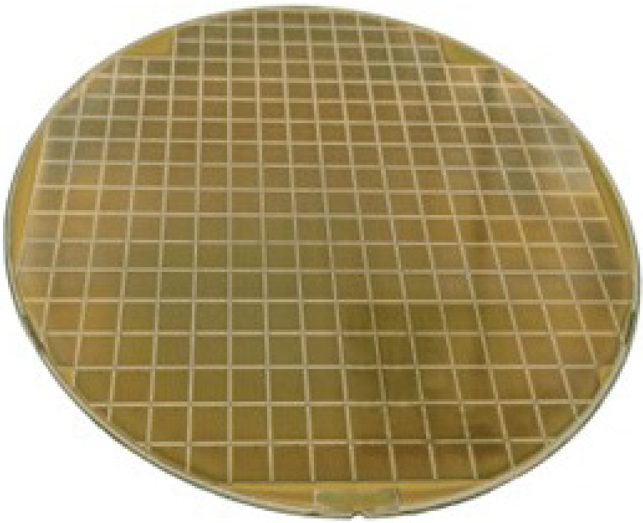
The Wiyo tag.

[2]Article 'From last option to better option', https://www.imeciclink.com/en/articles/last-option-better-option

[3]Article 'A start-up doesn't have the time or money to make mistakes', https://www.imeciclink.com/en/articles/start-doesnt-have-time-or-money-make-mistakes

Microprocessors for space applications

Frontgrade Gaisler develops radiation-hardened and fault-tolerant microprocessors for space, enabling groundbreaking scientific missions, such as the GR718B SpaceWire router currently en route to Jupiter. ASICs are a logical choice because of the **demanding requirements of space applications**, where performance, reliability, and radiation tolerance are paramount.[4]



Wafer of Frontgrade Gaisler's GR718B, processed on UMC 180nm with imec's DARE180 radiation-hard library.

Secure CPUs to withstand cyber threats

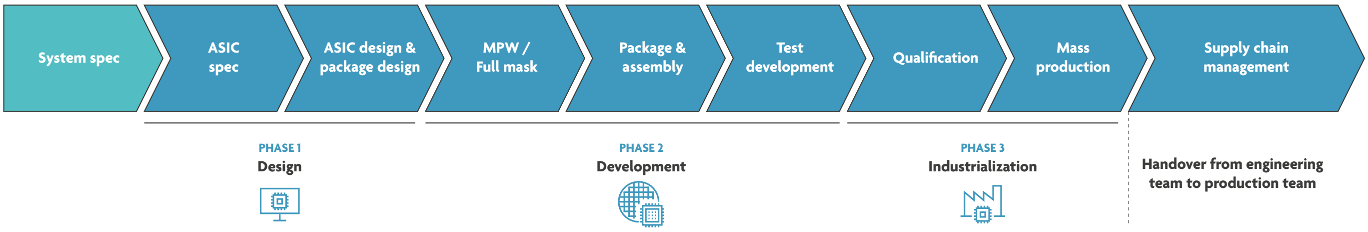
Arm is a global leader in semiconductor IP. For the Morello research program they needed to create a security-optimized CPU architecture. They designed a prototype system-on-chip (SoC) capable of evaluating the impact of security-based architectural innovations on hardware and software. The system-on-chip design was complex with four CPUs, an 18-layer substrate, and compatibility with diverse workloads ranging from smartphones to data centers. ASIC technology was essential to meet the **high-performance requirements**, ensure versatility, and deliver a real-life prototype capable of rigorous evaluation by industry and academia.[5]

Morello SoC on a board.

[4]Article 'A shared journey: Frontgrade Gaisler and IC-Link', https://www.imeciclink.com/en/articles/shared-journey-frontgrade-gaisler-and-imecic-link
[5]Article 'Collaborating for a secure future with Arm', https://www.imeciclink.com/en/articles/collaborating-for-a-secure-future-with-arm

The different stages in an ASIC journey

Designing and manufacturing an application-specific integrated circuit is a highly complex process that involves several distinct stages, each critical to ensuring the final product meets the desired performance, quality, and cost requirements. A company typically relies on **one or more partners** in this process because it requires a **wide range of specialized skills**. Many of these require a high level of expertise and must be continuously updated and refined, given the rapid pace of technological change in the semiconductor industry.



Custom ASIC flow.

Weighing chip options

The first alignment meeting will focus on a high-level discussion of the product or application, including required functionalities, performance targets, cost constraints, power and area requirements, and more. Additionally, the team will assess whether an ASIC is the best option, considering **expected volumes** and **time-to-market** goals. It is also important at this time to review the **existing skillsets** within the customer's company.

Translating product requirements into system architecture

Once it is decided that an ASIC is the best option for the product or application, the **system requirements** are listed. These requirements outline the system's deliverables and define the framework within which the ASIC must function. System requirements guide the development of the ASIC by **providing context and boundary conditions** for its design.

For example:

- The primary functions the system must perform (e.g., signal processing, data encryption, image processing);
- Interaction with other components in the system, such as sensors, memory, or external processors;
- Speed: maximum clock frequency, latency, or throughput requirements;
- Power: constraints on power consumption (critical for battery-operated or energy-efficient systems);
- Accuracy: requirements for precision in computations or data handling;
- Operating temperature range, humidity, and other physical conditions the system will encounter like vibration, shock, or radiation.

Also, a decision needs to be taken on what part of the system will **run in software** and what will be **implemented in hardware**. High-level **modelling and simulation** is performed so that the whole system can be proven to meet the requirements. Think of mathematical concepts like algorithms, equations, impulse responses, and finite-state machines.

Defining the ASIC specifications

The next step in the process is **translating** the system hardware requirements into ASIC specifications. It involves breaking down the high-level requirements of the overall system into **detailed, technical specifications** that define how the ASIC should be designed and what it must achieve. Decisions are made on which parts of the chip will be **analog**, which ones will be **digital**, which process **technology node**. Plus, the specs will include how it will be **packaged and tested in production**. This involves balancing performance, power and cost. You could consider the ASIC specifications as a very early datasheet of the chip.

During this phase, decisions are made on the use of **(design) IP**, which is pre-designed functionality. There are different kinds of IP, at different levels of maturity. Design IP includes digital standard cell and IO libraries, on-chip memories, pre-designed interface circuitry, processors, busses, SERDES, PCIe interfaces, memory controllers, PLLs etc. Not having to design everything from scratch does **decrease the development time** considerably.

In this phase also a **functional verification plan** is created, that will be used throughout the ASIC design phase to verify whether the design is living up to the specifications.

And also, a **test plan** is made. The test plan is defining how the chip will be adapted to make it production testable. This test plan is not to test the functionality as such, but to test manufactured chips against fabrications flaws.

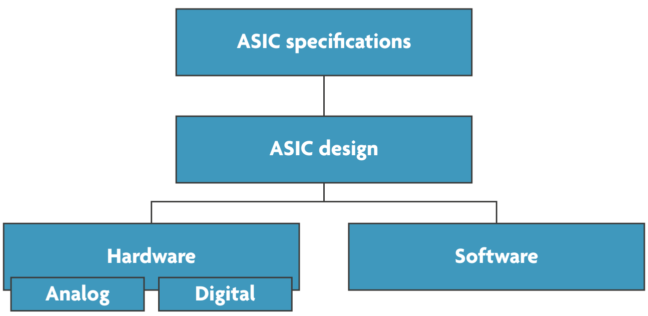
ASIC design

ASIC design can be divided into two parts:

- **The functional design** defines **how** the ASIC should perform its intended tasks or operations according to the system's requirements. In other words, the functional design specifies what the chip does – its core functionality – and how it interacts with inputs, processes data, and produces outputs. Parts of this functionality will be **analog circuitry**; other parts will be **digital**. At this stage certain design IPs (if used) will be integrated into the overall design. The analog circuitry is designed 'per transistor', while the digital design is done in RTL (Register Transfer Language). The largest part of functional design is creating **verification test benches** supporting the verification plan and proving the functional correctness of the chip. Digital and analog design are two very different fields, with **different tooling**. There are in fact even sub-specializations in both fields (RF, MMWAVE, high-speed analog etc.). In this phase, a **power intent file** is developed along with a **constraints file** (power, area, timing). Both are important to make sure that the implementation in the next step is resulting in a functionally correct design that runs at the required speed and supports the correct power modes.
- The **non-functional design (or implementation)** encompasses **DFT (design for test)** and **PD (physical design)**. DFT is using a set of models, techniques and methodologies to make manufactured chips **easily testable against manufacturing defects**. If a part of the chip is badly testable, there will be test escapes – bad chips perceived as good ones – which will lead to problems in the field and loss of revenue due to having to replace products. Physical design is the translation of the functional description of the chip into a **manufacturable layout** for a given process technology node. Physical implementation for the analog parts is called 'full custom layout' and needs skilled transistor-level layouters using a specific tooling. For digital circuits this step is referred to with terms like digital layout, APR (automatic place & route) or simply RTL to GDS (referring to the file format sent to fabrication). Digital layout requires a highly skilled team and expensive software tools to create and verify the layout. This ensures the design is manufacturable, functions correctly, and meets reliability standards.

At every step in the chip design, verification is done. There are several methodologies, and which ones are to be used are defined in **the verification plan**. They differ for analog and digital circuits. Methodologies are e.g. analog, digital and mixed-signal simulation, emulation, AC-, DC- and transient analysis, formal verification, FPGA prototyping, assertion-based verification, hardware-assisted verification, coverage-driven verification, design rule, electrical rule, and layout vs schematic checks.

Note that the skills and the application knowledge needed of the **design team** largely depend on the specifications of the ASIC and are thus **very different from one product to another**. For example, designing an imager requires completely different skills than designing a power control chip. This is why it is sometimes difficult for a company to have all design expertise in house, and this can then be **complemented by an ASIC development partner**. Also, some design tools can be very expensive – like the ones for non-functional design – and should be considered in the calculation when deciding to do this in-house or to outsource.



Bridging ASIC specifications to design.

Package design

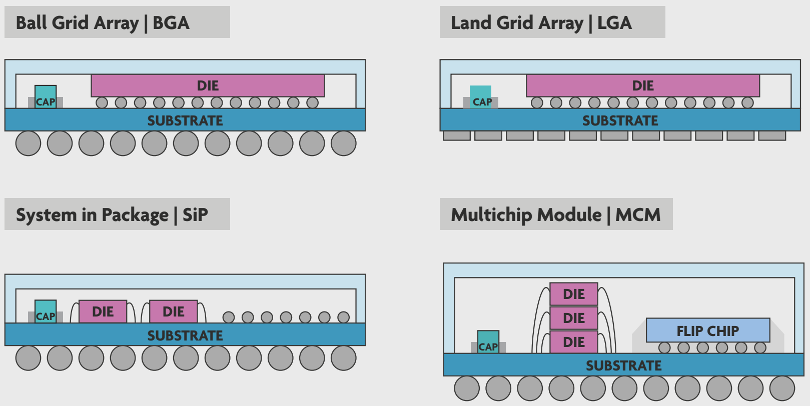
Package design goes hand in hand with the design of the ASIC. In the past, package design was often done after chip design, resulting in many failures. Indeed, a bad package design can kill a product even if the chip is designed perfectly. In recent times, emphasis is on **co-designing the ASIC and package** because the connectivity between the package and the chip is something that has to be tackled at both sides.

The package determines how the fabricated chip is enclosed, protected, and connected to the outside world. The package serves both as a mechanical enclosure and an electrical interface for the ASIC, and it must be carefully designed to meet the performance, reliability, and environmental requirements of the application. Based on the ASIC specifications and system requirements, the package size, type, and performance constraints of the package are defined.

There are different kinds of packages such as **wire-bond, flip-chip and wafer-level packages**, each with its own specific pros and cons. Note that the **targeted volumes** define the packaging options. Some options just aren't easily available for very low volumes (or are very expensive in that case).

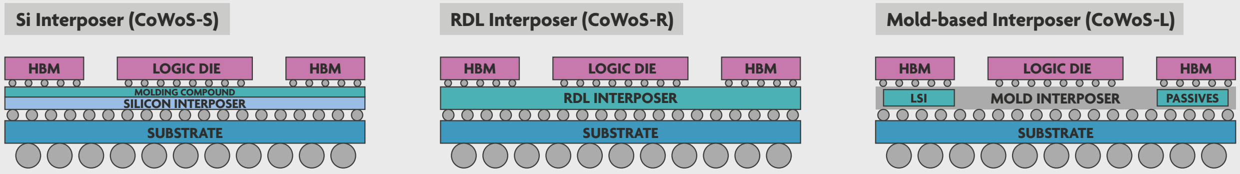
There is a wide variety of packaging techniques which needs to be carefully considered and again – just as the technology node selection – is the work of **a specialist team**. It is therefore difficult for a company to have all this knowledge in house and a partner such as IC-Link is key in the successful development of an ASIC.

Examples of mainstream packaging techniques

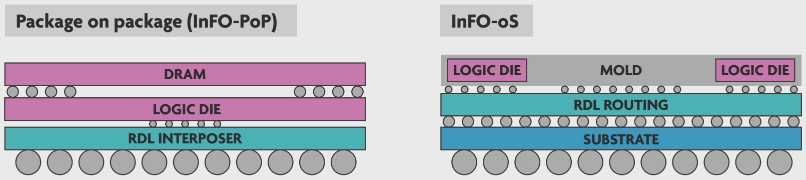


Examples of advanced packaging techniques: TSMC 3D Fabric™

Chip on Wafer on Substrate (CoWoS®)



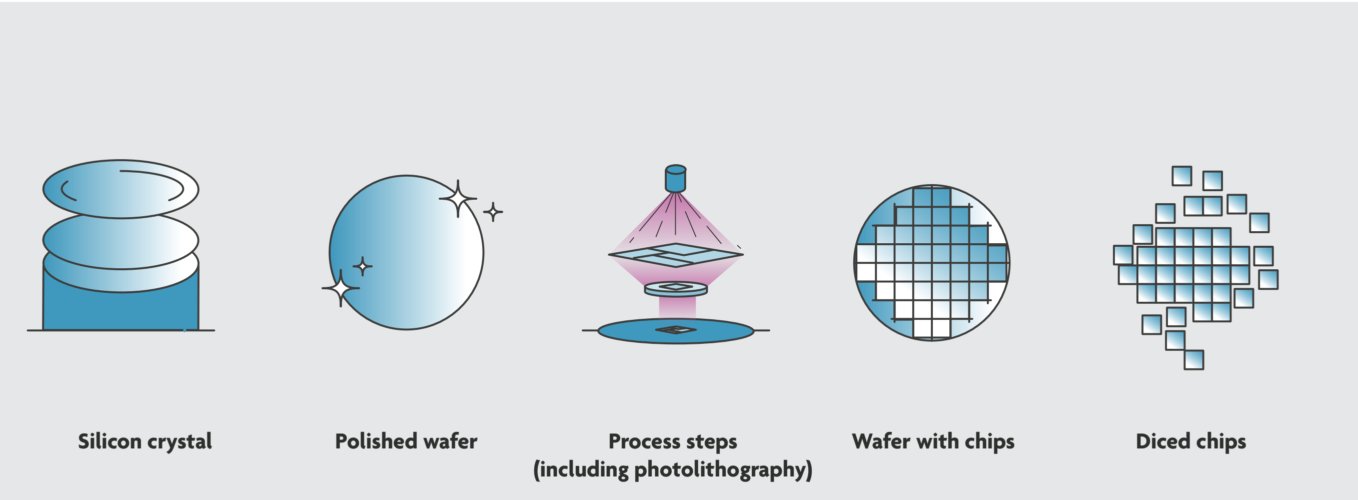
Integrated Fan-Out (InFO) (CoWoS®)



Mask development & manufacturing

Masks (also known as photomasks) are used to **define the patterns** on the silicon wafer during the **photolithography** process, which is a key part of chip manufacturing. These patterns define the circuit layout and ultimately determine how the ASIC functions.

Note that translating the design into manufacturing steps requires a **specialized team** that is only needed for a limited time. This is why it is interesting for a company to outsource this to an ASIC development partner.



Silicon manufacturing, from silicon wafer to diced chip. Wafer sizes have increased over the years, enabling to produce more semiconductor devices from a single wafer, enhancing productivity and efficiency. Also, the transistor size has evolved over the years, becoming smaller, and enabling faster switching and less energy consumption. It is referred to as a 'technology node'. Note that wafer sizes are linked to technology nodes and that the product or application defines which technology node is needed (= this decision is part of the ASIC design).

Package and assembly

Packaging is the process of enclosing the ASIC in a **protective shell** (to withstand environmental factors like heat, humidity, and mechanical stress) and creating the necessary **electrical connections** between the chip and the external environment.

Once the ASIC has been packaged, it needs to be assembled into the final product or system. This includes the process of placing the packaged ASIC onto a printed circuit board (PCB) or module and connecting it to other components.

Test development

Testing can be broadly categorized into two types: **functional (bench) testing** and **production testing**. Both are important but serve different purposes in the overall process.

Functional testing primarily focuses on verifying that the ASIC performs the intended tasks correctly. It is often done using a 'bench' setup, where the chip is tested under controlled conditions to check whether it meets the **desired functional specifications**. In this type of testing, the chip is typically assessed in **real-world scenarios**, ensuring it operates as expected. Functional testing is also done over temperature and voltage corners provided by the ASIC specification.

Production testing, on the other hand, is more focused on ensuring that the chip is **free of defects** and can be **reliably manufactured at scale**. It verifies that the chip meets the manufacturing process specifications and identifies defects that might not be caught by functional testing. Manufacturing testing is typically the more critical phase for **ensuring profitability** in the production of ASICs because it helps to identify any defects or faults early in the production process. This phase is often more automated and involves techniques like DFT (design for testability) and ATPG (automatic test pattern generation), which help to test the chip's electrical integrity and functionality under specific conditions.

Qualification

In the ASIC development process, qualification refers to a series of rigorous tests and evaluations to ensure that the ASIC meets the **required performance, reliability, and environmental standards** for its intended application and lifecycle. It is a crucial step for confirming that the ASIC can perform consistently and reliably under real-world conditions and that it adheres to the specifications outlined during the design phase.

Mass production

The mass production phase is the **final stage** after successful testing and qualification. This stage involves the **full-scale fabrication** of chips using the finalized design and manufacturing process, with all design adjustments and improvements already incorporated.

There are 2 phases in this production stage:

- **The engineering phase** is the initial production phase, typically encompassing the **first 300 wafers**, and focussing on gathering data and learning about the chip's behavior under real manufacturing conditions (characterization) and optimizing the test process. This step is also referred to as **industrialization**.

- **The supply phase** starts once the engineering phase yields a stable and optimized process and focuses on scaling up production and ensuring a **smooth ramp-up** to meet market demand.

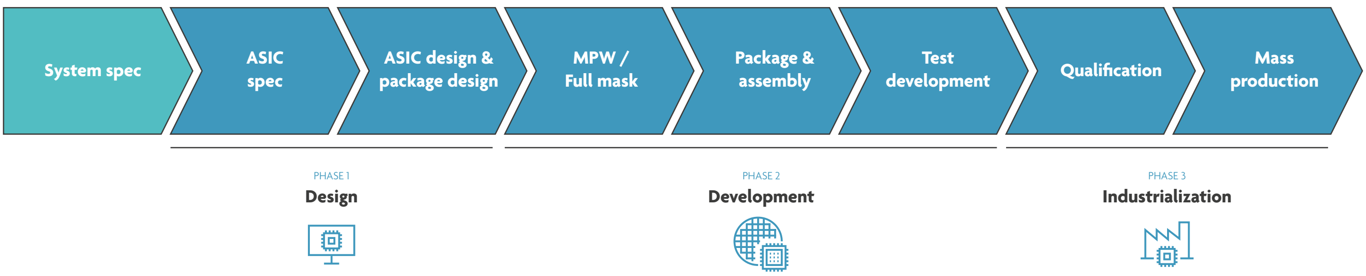
Supply-chain management is crucial in the mass production phase. Effective management ensures that materials, components, and resources are available at the right time, and helps prevent delays or disruptions in the manufacturing process.

Introducing IC-Link as your ASIC partner

IC-Link has its roots in **imec**, a world-renowned **research and innovation hub** in nanoelectronics and digital technologies. Established in 1984, imec initially focused on cutting-edge research in microelectronics and semiconductor technology. Over the years, it evolved into a leading partner for companies and research institutions looking to develop innovative solutions in areas such as healthcare, energy, mobility, and information technology.

Recognizing the increasing demand for specialized expertise in the field of ASICs, imec launched IC-Link as its **dedicated chip manufacturing division**. This strategic move aimed to **bridge the gap between advanced research and practical, market-ready products**. IC-Link officially began operations in 2013, leveraging imec's extensive research capabilities, cutting-edge technologies, and partnerships within the semiconductor ecosystem.

From its inception, IC-Link has focused on providing comprehensive support to clients throughout the **entire ASIC lifecycle**. This encompasses design services, access to advanced manufacturing capabilities for prototyping and volume, package design, assembly and supply chain services including production engineering and, if needed, failure analysis. By offering **tailored solutions**, IC-Link has helped numerous companies from various industries successfully transition from concept to production, ensuring that their custom ASICs meet the highest standards of quality and performance.



Flexible business models

BUSINESS MODEL 1

FULL TURNKEY

Phase 1, 2 & 3: imec as **one single point of contact** all the way through the development.

BUSINESS MODEL 2

CUSTOMER-OWNED TOOLING

"Pick & choose" type of offering: probably the **lowest cost approach** but cusomer takes the yield risk.

IC-Link has a flexible way of working, tailored to the needs of the customer, with the full turnkey (FTK) model being more low risk and the customer-owned tooling (COT) model more low cost. More info here.

Typical reasons why companies contact IC-Link

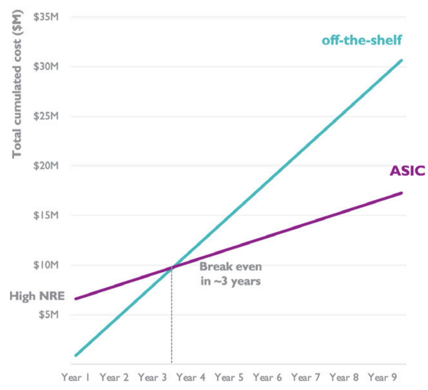
Here are some typical questions that IC-Link receives from companies that are considering an ASIC solution. They can serve as inspiration to help determine whether this approach is a good fit for your business case.

- ***A component in my PCB-based system has become obsolete / is not readily available. Maybe a good moment to consider turning the board into an ASIC?***
  Yes, an ASIC will give you more control over the supply chain and uses the technology that is tailored to your product's specifications, but actually it is so much more. It allows you to add unique functionalities to your product and to differentiate it from those of your competitors. So, instead of 'just' translating the board functionality into an ASIC, let's take one step back and consider all wanted specifications from the design phase onwards.

- ***Can IC-Link provide us with a specific technology or packaging solution and what is the price?***
  Thanks to our extensive ecosystem and the multitude of expertise available at IC-Link, you get access to established and new technology options. Yet, it is interesting to first discuss which options are available, with their pros and cons. As such, IC-Link can advise you on what the best technology option is for your product.

- ***I want to go to high volumes.***
  ASICs are a good idea when you want to move to high volumes because it will benefit the costs of your product. As an example, let's consider a 7nm processor ASIC for high-performance computing in datacentres. The total non-recurring engineering cost would approximately be 40 million USD. The unit cost of the ASIC would be in the range of 50 - 60 USD for a given die size, compared to a similar off-the-shelf CPU priced at 90 – 100 USD. For an annual production volume of 250,000 units and a 9-year product lifecycle, the ROI of the ASIC is estimated to be x1.26. The break-even point is projected to be reached within 4-5 years.

Cost category	Cost ($K)
Design	20,000
IP (3rd party)	8,000
Mask	9,000
Engineering lot	600
Package development	500
Production test development	1,300
Qualification	500
Total NRE	**39,900**



- ***I'm entering a new market and my current solution is too big and not reliable enough.***
  An ASIC solution is much more compact and performs better than a solution with standard components. Also, because there are fewer connections between the different components, it's easier to test the ASIC and the reliability and robustness is higher.

- **I'm designing my first prototype and want to take into account a future ASIC implementation.**
  An ASIC solution is not often considered when designing the first prototype. Reasons are of course the large investment, both in terms of cost and time. However, it's extremely interesting to think about an ASIC approach from the beginning, e.g. in your choice of components. It will make the future transition from a standard solution, like an FPGA, to an ASIC, significantly easier. For example, FPGA environments offer a wide range of interfaces and tools to get a system up and running quickly. However, when transitioning to an ASIC, challenges can arise. Suppose your design relies on a high-speed interface: purchasing the IP for that interface in an ASIC could cost hundreds of thousands or even millions of dollars. Alternatively, if you had chosen a simpler interface, like an SPI (serial peripheral interface), it might have required only a bit of coding without additional costs. Similarly, if you select a processor tied specifically to the FPGA that isn't widely available elsewhere, it can complicate the future ASIC design. Opting instead for a more universal processor, such as an ARM core, would avoid this issue. So, by considering the ASIC path upfront, you can save costs, reduce complexity, and streamline the development process.

ASIC projects to illustrate the role of IC-Link

To highlight the different roles that IC-Link can take up in an ASIC project, we present here some specifics on the collaboration with the companies making the migraine-curing implant, smart tags, microprocessors for space and secure CPUs, as described on page 6-7.

Access to advanced node (7nm)

The Six Semiconductor develops DDR PHY solutions: physical interface (PHY) layers to enable communication between a memory controller and DDR (double data rate) memory, often used in computers, servers, mobile devices, and embedded systems for high-speed memory access. For the development of their two 7nm memory sub-system chips, they partnered with IC-Link because of the technical and logistic complexity of such an advanced technology node. Also, they needed an advanced packaging technique (3D) and it is difficult for a relatively small company to get access to such a technology in a large chip foundry.

Technology advisor

Wiyo is a Spanish start-up that produces smart tags, to track or interact with all kinds of objects, e.g. in hospitals, stores, industrial environments etc. IC-Link acted as their advisor to get a complete overview of semiconductor technologies, and to explain the importance of making the right technology choices from the beginning of the design phase. Finally, a TSMC technology was chosen, on a multi-project wafer. Currently, the company is in the phase of proof-of-concept design and production for industrial testing, piloting, and calibration.

Reliable access to foundries

Frontgrade Gaisler develops radiation-hardened and fault-tolerant microprocessors for space, enabling groundbreaking scientific missions. The company has worked with IC-Link since 2004, for access to imec's DARE radiation-hard design platform, for support in creating specific IP and addressing mixed-signal design challenges for their ASICs, for affordable prototyping through multi-project wafer runs and reliable access to foundries for manufacturing final flight models.

End-to-end support, and knowhow on complex packaging

Arm is a global leader in semiconductor IP. For the Morello research program they needed to create a security-optimized CPU architecture. To fabricate the Morello SoC, Arm partnered with IC-Link to provide end-to-end support across the supply chain. IC-Link managed the complex SoC packaging process, working with Kyocera to produce the intricate 18-layer substrate. They also facilitated a TSMC full mask set tape-out and supported critical design validations, including a focused ion beam (FIB) process to implement a design fix efficiently.

From design to manufacturing, in 180nm & 55nm technology

Capri-Medical is an Irish start-up that develops a miniature bioelectronics device that is implanted in the back of the neck, to stimulate the peripheral nervous system and cure chronic migraine. Together with IC-Link, the company designed and developed an ASIC, with a first design in TSMC 180nm technology, on a multi-project wafer. This mature technology was chosen as a starting point since it was the most cost-effective for the first prototype. The plan is to further shrink the ASIC by moving to smaller technology nodes, such as 55nm.

Open communication and trust are the foundation of every successful ASIC project because you must be able to work closely with experts, sharing your vision for future products and defining the precise specifications for your current design. True success comes from collaboration – bringing together diverse expertise and perspectives to innovate as a unified team.

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